

Alireza Salemi

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RESEARCH SUMMARY

PhD candidate and Google PhD Fellow working at the intersection of information retrieval and large language models, with research experience at Google DeepMind and Google Cloud AI.

Interests: RAG · LLM personalization · Post-training · Alignment · Agentic systems · Reasoning

EDUCATION

University of Massachusetts Amherst

Sep 2022 – Present

PhD, Computer Science (MSc 2024) · GPA 3.93/4.0 · advised by Hamed Zamani

- Awarded the **Google PhD Fellowship 2025** in Natural Language Processing.
- Two **SIGIR best-paper awards**: Best Short Paper (2024) and Best Student Paper (2023).
- **2nd place** (\$30K) at the 2026 Grounded Reasoning Cup, an AI-agent competition sponsored by Google, OpenAI, and Anthropic, ahead of teams from top universities.

University of Tehran

Sep 2017 – Feb 2022

BSc, Computer Engineering · GPA 19.60/20 (ranked 1st of 103) · advised by Azadeh Shakery

RESEARCH EXPERIENCE

Student Researcher — Google Cloud AI

Jun 2025 – Dec 2025

Hosts: Hamid Palangi and Mihir Parmar

- Designed blackboard-style multi-agent systems for autonomous information discovery in data-science workflows; 13–57% relative gains over the strongest baselines.

Student Researcher — Google DeepMind

Jun 2024 – May 2025

Hosts: Cheng Li, Mingyang Zhang, and Michael Bendersky

- Personalized and aligned LLMs with reinforcement learning, reinforced self-training, inference-compute scaling, reasoning, and pluralistic alignment (REST-PG, Pathways of Thoughts).

Data Science Intern — Lowe's

May 2023 – Aug 2023

Host: Surya Kallumadi

- Optimized dense retrievers that feed personal documents to LLMs via reinforcement learning and knowledge distillation from the model's own feedback, improving LLM personalization.

Research Assistant — Center for Intelligent Information Retrieval

Sep 2022 – Present

Advisor: Hamed Zamani

- Research on RAG, LLM personalization, and agentic systems—building benchmarks, reinforcement-learning and self-training methods, and evaluation frameworks grounded in retrieved evidence.

NLP Data Scientist — Sotoon

Nov 2021 – Aug 2022

Team lead: Hakimeh Fadaei

- Built an AutoML system that selects models and hyperparameters for text classification, powering sentiment analysis and automatic moderation of user-generated product reviews.

PUBLICATIONS

📄 2,003 citations | h-index 17 | i10-index 18 | *Google Scholar, September 2026*

SELECTED PUBLICATIONS

LaMP: When Large Language Models Meet Personalization

ACL 2024

Alireza Salemi, Sheshera Mysore, Michael Bendersky, Hamed Zamani

[Paper](#) · [Code](#)

First benchmark for LLM personalization (7 tasks); 590+ citations and a standard reference for the field.

- Evaluating Retrieval Quality in Retrieval-Augmented Generation** *SIGIR 2024*
 Alireza Salemi, Hamed Zamani 🏆 **Best Short Paper** [Paper](#) · [Code](#)
eRAG scores each retrieved document by its downstream contribution—Kendall’s τ +0.17–0.49 at up to 50× lower GPU cost.
- A Symmetric Dual Encoding Dense Retrieval Framework for Knowledge-Intensive Visual Question Answering** *SIGIR 2023*
 Alireza Salemi, Juan Altmayer Pizzorno, Hamed Zamani 🏆 **Best Student Paper** [Paper](#) · [Code](#)
DEDR encodes queries and documents in a shared space; +11.6–30.9% retrieval and state-of-the-art end-to-end VQA.
- LLM-based Multi-Agent Blackboard System for Information Discovery in Data Science** *NeurIPS 2026*
 Alireza Salemi, Mihir Parmar, Palash Goyal, Yiwen Song, Jinsung Yoon, Hamed Zamani, Hamid Palangi, Tomas Pfister [Paper](#)
Specialized agents self-select tasks via a shared blackboard—no rigid hierarchy; 13–57% relative gains. (Google Cloud AI.)
- Optimization Methods for Personalizing Large Language Models through Retrieval Augmentation** *SIGIR 2024*
 Alireza Salemi, Surya Kallumadi, Hamed Zamani [Paper](#) · [Code](#)
First to optimize the retriever that feeds personal documents to an LLM via RL and knowledge distillation; significant gains on 6/7 LaMP tasks. (Lowe’s.)
- GrepSeek: Training Search Agents for Direct Corpus Interaction** *To be submitted (ICLR 2027)*
 Alireza Salemi, Chang Zeng, Atharva Nijasure, Jui-Hui Chung, Razieh Rahimi, Fernando Diaz, Hamed Zamani [Paper](#) · [Code](#)
Trains LLM agents to search corpora via shell commands using verified-trajectory generation + policy optimization (RL); strong gains across seven QA benchmarks.
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- OTHER PUBLICATIONS**
- Critic-R: Improving Agentic Search using Instruction-tuned Retrievers with Natural Language Introspective Feedback** *Findings of EMNLP 2026*
 Md Zarif Ul Alam, Alireza Salemi, Hamed Zamani [Paper](#)
- Improving User Privacy in Personalized Generation: Client-Side Retrieval-Augmented Modification of Server-Side Generated Speculations** *arXiv 2026*
 Alireza Salemi, Hamed Zamani [Paper](#) · [Code](#)
- Evaluation of Agents under Simulated AI Marketplace Dynamics** *SIGIR 2026*
 To Eun Kim, Alireza Salemi, Hamed Zamani, Fernando Diaz [Paper](#)
- Learning to Reason for Multi-Step Retrieval of Personal Context in Personalized Question Answering** *SIGIR 2026*
 Maryam Amirizani, Alireza Salemi, Hamed Zamani [Paper](#) · [Code](#)
- Learning from Natural Language Feedback for Personalized Question Answering** *SIGIR 2026*
 Alireza Salemi, Hamed Zamani [Paper](#)
- Beyond a Million Tokens: Benchmarking and Enhancing Long-Term Memory in LLMs** *ICLR 2026*
 Mohammad Tavakoli, Alireza Salemi, Carrie Ye, Mohamed Abdalla, Hamed Zamani, J. Ross Mitchell [Paper](#)
- Pathways of Thoughts: Multi-Directional Thinking for Long-Form Personalized Question Answering** *WWW 2026*
 Alireza Salemi, Cheng Li, Mingyang Zhang, Qiaozhu Mei, Zhuowan Li, Spurthi Amba Hombaiah, Weize Kong, Tao Chen, Hamed Zamani, Michael Bendersky [Paper](#)
- Plan-and-Refine: Diverse and Comprehensive Retrieval-Augmented Generation** *ICTIR 2026*
 Alireza Salemi, Chris Samarinas, Hamed Zamani [Paper](#) · [Code](#)
- LaMP-QA: A Benchmark for Personalized Long-Form Question Answering** *EMNLP 2025*
 Alireza Salemi, Hamed Zamani [Paper](#) · [Code](#)
- Personalized Generation in the Large Model Era: A Survey** *ACL 2025*
 Yiyan Xu, Jinghao Zhang, Alireza Salemi, Xinting Hu, Wenjie Wang, Fuli Feng, Hamed Zamani, Xiangnan He, Tat-Seng Chua [Paper](#)

ExPerT: Effective and Explainable Evaluation of Personalized Long-Form Text Generation	<i>Findings of ACL 2025</i>
Alireza Salemi, Julian Killingback, Hamed Zamani	Paper · Code
Beyond Factual Accuracy: Evaluating Coverage of Diverse Factual Information in Long-Form Text Generation	<i>Findings of ACL 2025</i>
Chris Samarin, Alexander Krubner, Alireza Salemi , Youngwoo Kim, Hamed Zamani	Paper · Code
Accelerating Retrieval-Augmented Generation	<i>ASPLOS 2025</i>
Derrick Quinn, Mohammad Nouri, Neel Patel, John Salihu, Alireza Salemi , Sukhan Lee, Hamed Zamani, Mohammad Alian	Paper
Reasoning-Enhanced Self-Training for Long-Form Personalized Text Generation	<i>arXiv 2025</i>
Alireza Salemi , Cheng Li, Mingyang Zhang, Qiaozhu Mei, Weize Kong, Tao Chen, Zhuowan Li, Michael Bendersky, Hamed Zamani	Paper
Learning to Rank for Multiple Retrieval-Augmented Models through Iterative Utility Maximization	<i>ICTIR 2025</i>
Alireza Salemi , Hamed Zamani	Paper
Comparing Retrieval-Augmentation and Parameter-Efficient Fine-Tuning for Privacy-Preserving Personalization of Large Language Models	<i>ICTIR 2025</i>
Alireza Salemi , Hamed Zamani	Paper · Code
CIIR@LiveRAG 2025: Optimizing Multi-Agent Retrieval-Augmented Generation through Self-Training	<i>LiveRAG @ SIGIR 2025</i>
Alireza Salemi , Mukta Maddipatla, Hamed Zamani	Paper · Code
Towards a Search Engine for Machines: Unified Ranking for Multiple Retrieval-Augmented Large Language Models	<i>SIGIR 2024</i>
Alireza Salemi , Hamed Zamani	Paper
LongLaMP: A Benchmark for Personalized Long-Form Text Generation	<i>Under review (TKDD)</i>
Ishita Kumar, Snigdha Viswanathan, Sushrita Yerra, Alireza Salemi , Ryan Rossi, Franck Dernoncourt, Hanieh Deilamsalehy, Xiang Chen, Ruiyi Zhang, Shubham Agarwal, Nedim Lipka, Hamed Zamani	Paper
Retrieval-Enhanced Machine Learning: Synthesis and Opportunities	<i>Under review (IR Journal)</i>
To Eun Kim, Alireza Salemi , Andrew Drozdov, Fernando Diaz, Hamed Zamani	Paper
Pre-Training Multi-Modal Dense Retrievers for Outside-Knowledge Visual Question Answering	<i>ICTIR 2023</i>
Alireza Salemi , Mahta Rafiee, Hamed Zamani	Paper
PEACH: Pre-Training Sequence-to-Sequence Multilingual Models for Translation with Semi-Supervised Pseudo-Parallel Document Generation	<i>LoResMT @ EACL 2023</i>
Alireza Salemi , Amirhossein Abaskohi, Sara Tavakoli, Yadollah Yaghoobzadeh, Azadeh Shakery	Paper · Code
ARMAN: Pre-training with Semantically Selecting and Reordering of Sentences for Persian Abstractive Summarization	<i>EMNLP 2021</i>
Alireza Salemi , Emad Kebriaei, Ghazal Neisi Minaei, Azadeh Shakery	Paper · Code
A Comparative Analysis of Toxic Span Detection using Attention-based, Named Entity Recognition, and Ensemble Models	<i>SemEval @ ACL 2021</i>
Alireza Salemi , Nazanin Sabri, Emad Kebriaei, Behnam Bahrak, Azadeh Shakery	Paper · Code

HONORS AND AWARDS

Grounded Reasoning Cup — 2nd Place and \$30K prize: live AI-agent competition on retrieval and reasoning over enterprise documents (Data + AI Summit 2026); sponsored by Google, OpenAI, and Anthropic; finished ahead of teams from leading universities.	2026
Google PhD Fellowship in Natural Language Processing: awarded to exceptional PhD students with outstanding research potential; proposal on collaborative optimization of agents.	2025
PhD Candidacy with Distinction: for exceptional performance in the PhD comprehensive exam.	2025

SIGIR 2024 Best Short Paper Award: Evaluating Retrieval Quality in Retrieval-Augmented Generation. 2024

SIGIR 2023 Best Student Paper Award: A Symmetric Dual Encoding Dense Retrieval Framework for Knowledge-Intensive Visual Question Answering. 2023

W. Bruce Croft Graduate Scholarship: for research in information retrieval and NLP. 2023, 2025

UMass Amherst CICS Scholarship: merit-based award for selected entering graduate students. 2022

Best Thesis Award, Computer Engineering: University of Tehran — BSc thesis selected as the best in the Computer Engineering graduating class for its research contribution. 2021

University of Tehran top-student awards: UT Sponsors Foundation Top Student, UT Scholarship, and Faculty of Engineering Award (top of 103 students). 2017–2020

TALKS

Agentic Search and Its Alternatives — Google Research Aug 2026

Personalization in the LLMs Era — Bocconi University May 2026

Personalization in the LLMs Era — Carnegie Mellon University Oct 2025

Personalization in the LLMs Era — Google DeepMind May 2025

Personalization in the LLMs Era — UMass Amherst Apr 2025

Knowledge-Intensive Visual Question Answering — QIAU Dec 2023

When Large Language Models Meet Personalization — UMass ML Club Oct 2023

When Large Language Models Meet Personalization — Google Research May 2023

PROFESSIONAL SERVICE

Organizing

- **Workshop organizer**, Second Workshop on Customizable NLP (CustomNLP4U) — ACL 2026.
- **Tutorial organizer**, Retrieval-Enhanced Machine Learning — SIGIR 2025 and SIGIR-AP 2024.

Program Committee and Reviewing

- **Program committee and reviewer** (80+ papers): NeurIPS, ACL, EMNLP, SIGIR, WWW, CIKM, ICTIR, SIGIR-AP, LLM-IGS; journals ACM TOIS and IEEE TKDE.

Mentoring

- Independent study — mentored 15+ students across RAG, personalization, and agents.
- Industry Mentorship Program — MS students on LLM personalization, with Adobe mentors.
- Early Research Scholar Program — undergraduates on QA and LLM personalization.